

# Universal Log Pre-processing Framework (ULPF)

Multi-Developer Work Breakdown Structure (WBS) & Phased Execution Roadmap

Doc Version: 1.0.0 • Baseline: Approved  
Target: ≥100k EPS • 100% Offline Air-Gapped

|                                          |                                                  |                                                     |
|------------------------------------------|--------------------------------------------------|-----------------------------------------------------|
| <b>Theme:</b> Blockchain & Cybersecurity | <b>Core Stack:</b> Rust Core + Python/Streamlit  | <b>Taxonomy:</b> OCSF v1.1.0 (Class 4001)           |
| <b>Target Throughput:</b> ≥ 100,000 EPS  | <b>Forensic Integrity:</b> Section 65B (SHA-256) | <b>Network Isolation:</b> 100% Air-Gapped / Offline |

## 1. ZERO-CONFLICT ARCHITECTURE & DEVELOPER ISOLATION RULES

To guarantee that multiple engineers develop in parallel without merge conflicts, broken builds, or blocking dependencies, the framework enforces four strict architectural barriers:

### The Four Conflict-Prevention Directives

- 1. Strict Folder Ownership:** Each engineer owns a designated root folder. No cross-directory edits are permitted.
- 2. Immutable Interface Contracts:** Boundary schemas (YAML rules, Parquet columns, Socket port 5140) are frozen before development.
- 3. Mock-Driven Autonomy:** Downstream tracks develop against mock data (e.g. Dashboard reads static Parquet samples while Engine builds the real-time sink).
- 4. Synchronous Handoff Protocol:** Completed tasks and interface updates are logged with author attribution into `conversation.md`.

## 2. INTERFACE CONTRACTS SPECIFICATION

| Interface Boundary                       | Producer & Consumer                                                        | Data Specification & Format                                                           | Endpoint / File Location                  |
|------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-------------------------------------------|
| <b>Contract A: Declarative Rules</b>     | <b>Producer:</b> Track 2 (Parsers)<br><b>Consumer:</b> Track 1 (Engine)    | Declarative YAML (signature match, regex groups, OCSF field aliases, disposition map) | <code>/parsers/*.yaml</code>              |
| <b>Contract B: Columnar Stream</b>       | <b>Producer:</b> Track 1 (Engine)<br><b>Consumer:</b> Track 3 (Dashboard)  | Snappy-compressed Apache Parquet & Arrow IPC batches conforming to OCSF Class 4001    | <code>/data/stream_buffer.parquet</code>  |
| <b>Contract C: Wire Log Intake</b>       | <b>Producer:</b> Track 4 (Test Tools)<br><b>Consumer:</b> Track 1 (Engine) | Raw UDP / TCP Syslog wire packets (RFC 3164 / 5424) pushed at high volume             | <code>0.0.0.0:5140</code>                 |
| <b>Contract D: Forensic Cryptography</b> | <b>Producer:</b> Track 1 (Engine)<br><b>Consumer:</b> Track 3 (Dashboard)  | Pre-parsing SHA-256 digest on raw bytes + RFC 4122 UUIDv4 bound into metadata         | <code>metadata.hash &amp; event_id</code> |

## 3. MODULAR TEAM TRACKS & ROLE ALLOCATION

|                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                               |
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| <div><b>TRACK 1</b> Core Ingestion Engine</div> <div>/core-engine/</div> <ul style="list-style-type: none"><li>Asynchronous UDP/TCP Socket Listener (Port 5140).</li><li>Hardware-accelerated SHA-256 raw byte hasher.</li><li>3-Tier Format Classifier &amp; OCSF Class 4001 Mapper.</li><li>Apache Arrow / Parquet high-speed batch sink writer.</li></ul>     | <div><b>TRACK 2</b> Declarative Parsers</div> <div>/parsers/ &amp; /data/geo/</div> <ul style="list-style-type: none"><li>YAML rules for Cisco ASA, Palo Alto, Fortinet, Check Point.</li><li>Offline MaxMind GeoIP integration (<code>GeoLite2-City.mmbd</code>).</li><li>Dynamic hot-reload schema validator (<code>validate_parsers.py</code>).</li><li>Static OCSF dictionary lookup tables.</li></ul>    |
| <div><b>TRACK 3</b> Dashboard &amp; Forensics</div> <div>/dashboard/</div> <ul style="list-style-type: none"><li>Streamlit Real-Time Raw-to-OCSF Split-Screen.</li><li>Section 65B 1-Click SHA-256 Forensic Integrity Verifier.</li><li>Interactive Drag-and-Drop YAML Parser Manager.</li><li>AI Threat Visualizer (Isolation Forest Anomaly Hunter).</li></ul> | <div><b>TRACK 4</b> Traffic Generator &amp; Benchmarks</div> <div>/test_tools/ &amp; /sample_logs/</div> <ul style="list-style-type: none"><li>Multi-threaded UDP Syslog Generator (1k to 100k+ EPS).</li><li>Realistic multi-vendor firewall raw log corpus.</li><li>Benchmarking scripts measuring EPS, RAM, and P95 latency.</li><li>Stress-test anomaly &amp; malformed packet injection suite.</li></ul> |

Comprehensive 4-Phase Implementation Schedule

Detailed Milestone Matrix Across All Development Tracks

Phase 1 → Phase 4  
Zero-Conflict Timeline

| Phase & Goal                                  | Track 1: Core Engine                                                                                                                                                                                        | Track 2: Parsers & GeoIP                                                                                                                                                                                        | Track 3: Dashboard & Forensics                                                                                                                                                                                             | Track 4: Generator & Benchmarks                                                                                                                                                                                                  |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Phase 1</b><br>Scaffolding & Contracts     | <ul style="list-style-type: none"><li>Set up socket listener stub on port 5140.</li><li>Implement raw byte capture &amp; SHA-256 pre-parser hasher.</li><li>Create Parquet schema template.</li></ul>       | <ul style="list-style-type: none"><li>Write declarative YAML specs for Cisco ASA &amp; Fortinet.</li><li>Bundle offline GeoIP database.</li><li>Build regex syntax checker.</li></ul>                           | <ul style="list-style-type: none"><li>Scaffold Streamlit UI layout.</li><li>Implement mock Parquet data loader.</li><li>Build split-screen visual grid for incoming vs normalized logs.</li></ul>                          | <ul style="list-style-type: none"><li>Assemble sample raw log files for 5 vendors.</li><li>Build basic multi-vendor UDP packet sender.</li><li>Verify network packet delivery to port 5140.</li></ul>                            |
| <b>Phase 2</b><br>Core Feature Implementation | <ul style="list-style-type: none"><li>Implement 3-tier detector (Vendor → Structure → Regex).</li><li>Build YAML parser execution engine.</li><li>Implement OCSF Class 4001 field mapping engine.</li></ul> | <ul style="list-style-type: none"><li>Complete Palo Alto, Check Point &amp; pfSense YAML specs.</li><li>Validate token extraction against 1,000+ raw test logs.</li><li>Map vendor disposition codes.</li></ul> | <ul style="list-style-type: none"><li>Implement Section 65B 1-click SHA-256 cryptographic verifier.</li><li>Build live parser upload / editor UI.</li><li>Implement Isolation Forest unsupervised anomaly model.</li></ul> | <ul style="list-style-type: none"><li>Upgrade traffic generator to multi-threaded async engine.</li><li>Build burst and rate-throttled modes (1k to 100k EPS).</li><li>Add malformed &amp; evasion log attack vectors.</li></ul> |
| <b>Phase 3</b><br>Integration & Pipeline Sync | <ul style="list-style-type: none"><li>Stream live normalized batches into /data/stream_buffer.parquet.</li><li>Enable file watcher for zero-downtime YAML hot-reloading (&lt;50ms).</li></ul>               | <ul style="list-style-type: none"><li>Live hot-reload validation with active engine.</li><li>Stress-test parser fallback heuristics on unmapped fields.</li></ul>                                               | <ul style="list-style-type: none"><li>Connect dashboard to real-time Parquet stream.</li><li>Add live EPS counter, latency gauge, and active memory monitor.</li><li>Enable real-time alert toast notifications.</li></ul> | <ul style="list-style-type: none"><li>Run continuous 25,000 to 50,000 EPS multi-vendor stress stream.</li><li>Validate end-to-end delivery from wire socket to dashboard display.</li></ul>                                      |
| <b>Phase 4</b><br>100k+ EPS & Final Delivery  | <ul style="list-style-type: none"><li>Optimize memory footprint (≤60 MB target).</li><li>Minimize P95 processing latency (&lt;2.0 ms).</li><li>Enforce zero packet loss under sustained burst.</li></ul>    | <ul style="list-style-type: none"><li>Finalize edge-case regex rules and PII masking filters.</li><li>Lock schema documentation and parser authoring guide.</li></ul>                                           | <ul style="list-style-type: none"><li>Finalize executive styling and forensic audit report export.</li><li>Polish AI threat visualization scatter plots and risk heatmaps.</li></ul>                                       | <ul style="list-style-type: none"><li>Execute official 100,000+ EPS benchmark run.</li><li>Export benchmark charts, latency distributions, and test logs.</li><li>Verify 100% offline air-gap autonomy.</li></ul>                |

5. CROSS-TRACK INTEGRATION CHECKPOINTS

 Milestone Synchronization Gates

- Checkpoint A (End of Phase 1):** Verify mock data compatibility. Dashboard runs cleanly on Track 1's mock Parquet schema; Generator successfully sends test packets to Track 1's socket.
- Checkpoint B (End of Phase 2):** Track 2's YAML parsers parse Track 4's sample logs with 100% field accuracy; Track 3's Section 65B hash verifier successfully passes on mock records.
- Checkpoint C (End of Phase 3):** End-to-end pipeline operates live at ≥25k EPS without crashes, file locks, or memory leaks.

Detailed Action Plan: Phases 3 & 4 (Integration & Delivery)

Step-by-Step Execution Plan, Exit Criteria, and NTRO Deliverables Checklist

NTRO Evaluation Ready  
Defense & Forensic Standard

PHASE 3: End-to-End Pipeline Integration & Live Synchronization

INTEGRATION MILESTONE

Primary Objective: Transition from independent mocked modules to an active, synchronized, end-to-end data pipeline.

Step 3.1 • Parquet Stream Linkage

Track 1 configures the asynchronous writer to flush OCSF Class 4001 records into /data/stream\_buffer.parquet in micro-batches (e.g. 5,000 records/flush) without disk locking conflicts.

Step 3.2 • Hot-Reload Engine Hook

Track 2 drops a new parser rule (e.g. custom\_snort.yaml) into /parsers/ while Track 1's engine is actively receiving packets, verifying dynamic reload in <50ms without dropping packets.

Step 3.3 • Dashboard Live Feed Binding

Track 3 links the Streamlit dashboard to /data/stream\_buffer.parquet using PyArrow memory scanning, enabling live split-screen waterfalls and dynamic charts without UI lag.

Step 3.4 • 50k EPS Continuous Ingestion Test

Track 4 executes a sustained 10-minute traffic stream of mixed Cisco, Palo Alto, and Fortinet logs at 50,000 EPS. Verify 0 dropped events, accurate OCSF mapping, and unbroken SHA-256 bindings.

Phase 3 Exit Criteria: Complete ingestion-to-visualization cycle proven with 250,000+ events processed, zero crashes, and 100% forensic hash verification accuracy.

PHASE 4: 100k+ EPS Stress Benchmarking, Security Audit & Final Packaging

FINAL DELIVERY MILESTONE

Primary Objective: Prove defense-grade performance metrics, verify Section 65B legal defensibility, and package for air-gapped deployment.

Step 4.1 • 100,000+ EPS Benchmark Execution

Execute Track 4's high-speed benchmark harness across 4 CPU cores. Record throughput (target ≥100k EPS), P95 latency (target <2.0ms), and sustained memory usage (target ≤60MB RAM).

Step 4.2 • Section 65B Forensic Integrity Audit

Run automated cryptographic verification script across 100,000 stored records: re-compute SHA256(raw\_data) on every row and assert 100% bitwise identity against metadata.hash.

Step 4.3 • AI Threat Hunting Verification

Inject port-scan, data exfiltration, and brute-force attack vectors into the synthetic log stream. Verify that Track 3's Isolation Forest algorithm flags anomalies with score > 0.75.

Step 4.4 • 100% Air-Gapped Packaging

Verify that the system operates in a sandboxed offline environment with zero network adapters connected. Package single-command launch script and compile executive presentation.

Phase 4 Exit Criteria: All NTRO problem statement requirements verified; ≥100k EPS benchmark report produced; 1-click Section 65B integrity audit passing; 100% air-gap compliance verified.

6. NTRO EVALUATION DELIVERABLES & ARTIFACTS CHECKLIST

| Evaluation Deliverable           | Artifact Location                                | Responsible Track | Status          |
|----------------------------------|--------------------------------------------------|-------------------|-----------------|
| 1. Core Ingestion Engine         | /core-engine/ (Async socket, Hasher, Normalizer) | Track 1 (Lead)    | READY FOR BUILD |
| 2. Declarative Parsers & GeoIP   | /parsers/*.yaml & /data/geo/                     | Track 2 (Lead)    | READY FOR BUILD |
| 3. Forensic Web Dashboard        | /dashboard/ (Streamlit, Verifier, AI Model)      | Track 3 (Lead)    | READY FOR BUILD |
| 4. Benchmarks & Traffic Tools    | /test_tools/ (Generator, Benchmarks, Datasets)   | Track 4 (Lead)    | READY FOR BUILD |
| 5. Architecture & Security Specs | /docs/PRD.md, TECHNICAL_ARCHITECTURE.md          | All Tracks        | COMPLETED       |